

# Derry Tanti Wijaya

Associate Professor of Data Science · Computational Linguistics & Natural Language Processing

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|---------------------------------------------|---------------------------------------------------------|-------------------------------------|------------------------------------------|
| <b>4,388+</b><br>Citations (Google Scholar) | <b>\$2.3M+</b><br>Research Funding Raised<br>(PI/Co-PI) | <b>8</b><br>PhD Students Supervised | <b>90+</b><br>Peer-Reviewed Publications |
|---------------------------------------------|---------------------------------------------------------|-------------------------------------|------------------------------------------|

Publication venues include ICML, ICLR, and \*CL conferences. h-index: 28 (Google Scholar, 2026).

## PROFILE

Computer scientist and NLP researcher specializing in **multilingual, multimodal, and multicultural language technologies**. Ph.D. from Carnegie Mellon University (advisor: Tom Mitchell); previously Assistant Professor of Computer Science at Boston University and postdoctoral researcher at the University of Pennsylvania (advisor: Chris Callison-Burch). Research spans **mechanistic interpretability and steerability of language models, model editing, evaluation frameworks and LLM-as-a-judge, and data-efficient adaptation for low-resource languages**, with a parallel program building culturally grounded resources and benchmarks (indigenous scripts, honorifics, code-switching) and applying NLP to societal challenges including bias, polarization, toxicity, and misinformation. Author of work at **ICML, ICLR, ACL, NAACL, and EMNLP**; principal or co-investigator on grants from Google, DARPA, the Open Society Foundation, Danida, and others.

## RESEARCH INTERESTS

- **Interpretable & steerable LMs:** mechanistic interpretability of multilingual/multimodal models, safe model editing, controllability, hallucination reduction.
- **Evaluation of generative models:** meta-metrics, LLM-as-a-judge, rubric-agnostic reward reasoning, alignment with human preferences.
- **Low-resource & multilingual NLP:** data-efficient transfer (circuit-targeted fine-tuning), machine translation from dictionaries and grammars, code-switching generation.
- **Multicultural AI & cultural heritage:** culturally aware large multimodal models, preservation of indigenous scripts, honorific systems.
- **NLP for society:** bias, polarization, toxicity, hate speech, and misinformation; computational framing analysis; AI for health.

## ACADEMIC APPOINTMENTS

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| <b>Associate Professor, Data Science — Monash University, Indonesia</b><br>Program Coordinator, Data Science (2024–present). Co-Director, Monash Data & Democracy Research Hub (2023–present). | 2022 – present |
| <b>Assistant Professor, Computer Science — Boston University</b><br>Now Adjunct. Co-Director, AI and Education Initiative, Hariri Institute (2021–2022).                                       | 2018 – 2022    |
| <b>Postdoctoral Researcher, Computer &amp; Information Science — University of Pennsylvania</b><br>Advisor: Chris Callison-Burch.                                                              | 2016 – 2018    |

## EDUCATION

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| <b>Ph.D., Language Technologies Institute — Carnegie Mellon University</b><br>Advisor: Tom Mitchell. Thesis: VerbKB — A Knowledge Base of Verbs for Natural Language Understanding. | 2016 |
| <b>M.Sc., School of Computing — National University of Singapore</b><br>Advisor: Stephane Bressan.                                                                                  | 2010 |
| <b>B.Comp., School of Computing — National University of Singapore</b>                                                                                                              | 2004 |

## AWARDS & FELLOWSHIPS

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- **Best Data Researcher, Data Science Indonesia Award** (2021)
- **University Provost's Career Development Professorship, Boston University** (2019)
- **Anita Borg Scholarship, Google** (2013)
- **International Science & Technology Fellowship, Fulbright** (2010–2016)

## SELECTED PUBLICATIONS

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*Name in bold. See Google Scholar for the complete list of 60+ works (4,388+ citations, h-index 28).*

### Interpretability, Model Editing & Evaluation

- Do Language Models Track Entities Across State Changes?  
Z. Tang, Q. Zhao, G. Franco, **D. Wijaya**, A. Mueller, S. Schuster, N. Kim  
*ICML 2026* [\[link\]](#)
- Beyond Transfer Accuracy: Faithful Circuits for Controlled Low-Resource Adaptation  
K. Nur'aini, A. Purwarianti, A. F. Aji, **D. Wijaya**  
*Preprint (in submission)* [\[link\]](#)
- mR3: Multilingual Rubric-Agnostic Reward Reasoning Models  
D. Anugraha, S-Y. Hung, Z. Tang, A. E-S. Lee, **D. Wijaya**, G. I. Winata  
*ICLR 2026* [\[link\]](#)
- MetaMetrics: Calibrating Metrics for Generation Tasks Using Human Preferences  
G. I. Winata, D. Anugraha, L. Susanto, G. Kuwanto, **D. Wijaya**  
*ICLR 2025* [\[link\]](#)
- Deductive Closure Training of Language Models for Coherence, Accuracy, and Updatability  
A. F. Akyürek, E. Akyürek, L. Choshen, **D. Wijaya**, J. Andreas  
*ACL Findings 2024* [\[link\]](#)
- DUnE: Dataset for Unified Editing  
A. F. Akyürek, E. L. Pan, G. Kuwanto, **D. Wijaya**  
*EMNLP 2023* [\[link\]](#)
- RL4F: Generating Natural Language Feedback with Reinforcement Learning for Repairing Model Outputs  
A. F. Akyürek, E. Akyürek, A. Kalyan, P. Clark, **D. Wijaya**, N. Tandon  
*ACL 2023* [\[link\]](#)

### Multilingual, Multimodal & Multicultural NLP

- NusaAksara: A Multimodal and Multilingual Benchmark for Preserving Indonesian Indigenous Scripts  
M. F. Adilazuarda, M. Wijanarko, L. Susanto, K. Nur'aini, **D. Wijaya**, A. F. Aji  
*ACL 2025* [\[link\]](#)
- Do Language Models Understand Honorific Systems in Javanese?  
M. R. Farhansyah, I. Darmawan, A. Kusumawardhana, G. I. Winata, A. F. Aji, **D. Wijaya**  
*ACL 2025* [\[link\]](#)
- WorldCuisines: A Massive-Scale Benchmark for Multilingual and Multicultural Visual Question Answering on Global Cuisines  
G. I. Winata et al., **D. Wijaya**  
*NAACL 2025* [\[link\]](#)
- What Do Indonesians Really Need from Language Technology? A Nationwide Survey  
M. D. A. Kautsar, L. Susanto, **D. Wijaya**, F. Koto  
*EMNLP 2025* [\[link\]](#)
- Linguistics Theory Meets LLM: Code-Switched Text Generation via Equivalence Constrained Large Language Models  
G. Kuwanto, C. Agarwal, G. I. Winata, **D. Wijaya**  
*Workshop on Computational Developmental Linguistics 2026* [\[link\]](#)
- Mitigating Translationese in Low-resource Languages: The Storyboard Approach  
G. Kuwanto et al., **D. Wijaya**  
*LREC-COLING 2024* [\[link\]](#)
- Explain-then-Translate: An Analysis on Improving Program Translation with Self-Generated Explanations

Z. Tang, M. Agarwal, A. G. Shypula, B. Wang, **D. Wijaya**, J. Chen, Y. Kim  
*EMNLP Findings 2023* [\[link\]](#)

### NLP for Society: Bias, Polarization, Misinformation & Health

- A Multi-Labeled Dataset for Indonesian Discourse: Examining Toxicity, Polarization, and Demographics  
L. Susanto, M. Wijanarko, P. Pratama, Z. Tang, F. Akyas, T. Hong, I. Idris, A. F. Aji, **D. Wijaya**  
*ACL Findings 2025* [\[link\]](#)
- COVID-19 Vaccine Misinformation in Middle Income Countries  
J. Kim, B. R. Bak, A. Agrawal, J. Wu, V. J. Wirtz, T. Hong, **D. Wijaya**  
*EMNLP 2023* [\[link\]](#)
- Effects of #coronavirus Content Moderation on Misinformation and Anti-Asian Hate on Instagram  
T. Hong, Z. Tang, M. Lu, Y. Wang, J. Wu, **D. Wijaya**  
*New Media & Society 2023* [\[link\]](#)
- Agenda Setting, Cross-Cutting Effects, and Political Expression on Social Media: The Gun Violence Case  
L. Guo, Y. Zhang, K. Mays, A. F. Akyürek, **D. Wijaya**, M. Betke  
*Communication Research 2023* [\[link\]](#)
- Proposing an Open-Sourced Tool for Computational Framing Analysis of Multilingual Data  
L. Guo, C. Su, S. Paik, V. Bhatia, V. P. Akavoor, G. Gao, M. Betke, **D. Wijaya**  
*Digital Journalism 2023* [\[link\]](#)

### COMPETITIVE RESEARCH GRANTS

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Over \$2.3M USD raised as Principal Investigator or Co-Investigator. Selected:

- **IndigiSEA: Indigenous Digital Futures for Southeast Asia** — Monash Incubator Grant, \$53k, 2026
- **Digitalising Waste Management for a Just Green Transition in Indonesian Smart Cities** — Danida (Denmark), \$300k, 2025-2029
- **Sea-Tech: Sustainable Seaweed Cultivation through Smart Technology** — PAIR (Australia-Indonesia), \$61k, 2025-2027
- **Contextualizing LLMs for Women's Sexual & Reproductive Health in Indonesia** — Google Research Award, \$60k (PI), 2024
- **ASEAN Disinformation Index** — Monash Data Futures Institute, \$33k, 2024
- **Boosting Monash's Global Reputation: Data & Democracy Hub** — Monash Incubator Grant, \$53k, 2024
- **Monitoring Hate Speech toward Vulnerable Communities, 2024 Indonesia Election** — Open Society Foundation, \$37k, 2023
- **Exploring Generative AI for Moderating Online Polarisation** — Monash Data Futures Institute, \$35k (PI), 2023
- **Bhinneka: Multilingual Indonesian Resources** — Indonesian Ministry of Education (US Collaboration), \$45k, 2022-2023
- **Exploring the Evolution of Racial Biases through Framing Analysis** — Google Research Scholar Award, \$60k (PI), 2021
- **LEARN: Label-Efficient Active Resilient Network** — DARPA (Learning with Less Labels), \$462k (Co-PI), 2019-2022
- **Bridging Linguistic and Visual Knowledge through Visual Genome** — Hariri Institute Research Incubation Award, \$28k (PI), 2019-2021
- **AIDA: Semi-supervised Learning of Multimodal Representations** — DARPA, \$100k (with C. Callison-Burch), 2019-2020
- **ExploreCSR: Supporting and Promoting Undergraduate Research for Women** — Google, \$18k, 2019
- **BIGDATA: Multiplatform, Multilingual, and Multimodal Tools for Analyzing Public Communication in 100+ Languages** — NSF, \$1,000,000 (with M. Betke, L. Guo, P. Ishwar), 2018-2022

### RESEARCH SUPERVISION

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#### Doctoral Students (8 total: 4 ongoing, 4 completed, all with top-venue output)

- **Ongoing:** Khumaisa Nur'aini (Monash), Adryan Kusumawardhana (Monash), Zilu Tang (BU), Ge Gao (BU).
- **Tahsina Hashem** (Monash, 2026) — Faithful and Fair Text Generation from Structured and Visual Data.
- **Muhammed Yusuf Kocyigit** (BU, 2025) — Improving Evaluation Practices in NLP: Contamination and Metric Limitations.

- **Afra Feyza Akyürek** (BU, 2025) — Advancing Neural Networks Beyond Initial Training.
- **Isidora Tourni** (BU, 2023) — Neural Machine Translation for Low-Resource Conditions.

#### Master's Students (selected; many with first-author papers at ACL/ICLR)

- **Garry Kuwanto** (BU, 2026) — MetaMetrics: a Bayesian optimization framework for calibrating LLM evaluation metrics (ICLR 2025).
- **Iwan Darmawan** (Monash, 2024) — Javanese Honorific System corpus (ACL 2025).
- **Fariz Akyas** (Monash, 2024) — Annotator Identity in Indonesian Political Polarization Corpus (ACL Findings 2025).

*Full supervision record (15+ Master's and Undergraduate students) available on request.*

## TEACHING

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Unit Coordinator, Master of Data Science, Monash University Indonesia. Selected courses with most recent student evaluation (SETU) scores:

- **Machine Learning (ITI5201)** — 82/100 (2026). Grounds ML in real-world societal challenges.
- **Data Analysis for Semi-Structured Data (ITI5212)** — 86/100 (2026). Focus on generative AI and LLMs.
- **Master Thesis supervision (ITI5126–5229)** — 88/100 (2025). Students published and presented at top conferences.
- **Data Wrangling (ITI5196)** — 86/100 (2024)
- **Introduction to Data Science (ITI5145)** — 82/100 (2022)

## LEADERSHIP & PROFESSIONAL SERVICE

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### Academic Leadership

- **Program Coordinator, Data Science**, Monash University Indonesia (2024–present).
- **Co-Director, Monash Data & Democracy Research Hub** (2023–present) — partnerships across government, academia, and civil society; convened the Data & Democracy Dialogue (Jakarta 2024; Kuala Lumpur 2025).
- **Faculty of IT Course Governance, Quality & Resourcing Sub-committee** (2026–present); Education Committee (2024–present); Research Committee (2024–2025).
- **Co-Director, AI & Education Initiative**, Hariri Institute, Boston University (2021–2022).

### Community Service to the Field

- **Area Chair / Senior Area Chair / Session Chair** for ACL conferences, ICLR, and NeurIPS (ongoing).
- **Organizer:** Demo Chair AACL 2026 (Hengqin); 2nd Workshop on Southeast Asian Language Processing (COLING 2025); 7th Workshop on Computational Approaches to Linguistic Code-Switching (NAACL 2025); Local Chair AACL 2023 (Bali).

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*References available upon request.*